

HealthConnect Clinic

Week 7 Project Summary — Data Analytics Track

1. What I planned to test

Whether the Week 6 statistical outputs (driver validation, the compound risk score, and the reminder-by-risk-tier finding) were correct and reproducible, whether the risk-score gradient held across patient segments, and whether a more powerful statistical test would change the Week 6 "suggestive" reminder-by-risk conclusion.

2. What I actually tested

Reproduced every Week 6 KPI and statistic from the raw dataset; audited the dataset for data-quality issues; tested the risk-score gradient across gender, age group and appointment type; replaced Week 6's two split-sample reminder tests with a single, more powerful pooled interaction model; and validated the risk score against an independent model-based benchmark.

3. Most important testing results

All Week 6 statistics reproduced exactly except one: the reported "risk score ≥ 2 " no-show rate (59.0%) was an unweighted average rather than the correct n-weighted pooled rate (56.0%). The risk-score gradient held in 12 of 13 tested segments. The reminder-by-risk finding, retested with a properly powered interaction model on the full sample ($p = 0.732$), is now confirmed rather than merely suggestive.

4. Issues or weaknesses identified

An unweighted-average error in one headline KPI; an under-powered split-sample test used in Week 6 for the reminder-by-risk question; and (initially) an unresolved cross-track dependency, since Data Science's Week 6 model output had not been received in time to fully validate the risk score against it.

5. Improvements / refinements made

Corrected the headline no-show rate to 56.0%; replaced the split-sample reminder test with a single pooled interaction model; rebuilt the dashboard to show the correction and the new tests; closed the cross-track dependency by incorporating Data Science's actual Week 7 refined model and held-out benchmark once received.

6. Retesting results

The corrected 56.0% figure, the segment-stability check, and the pooled interaction test all reproduce consistently on repeated runs of the notebook.

7-9. Cross-track collaboration

Track collaborated with: Data Science. **What was tested collaboratively:** the compound risk score was compared against Data Science's actual Week 7 refined model (a regularized Logistic Regression, 5 features), evaluated on their held-out test set. **What changed as a result:** the original in-sample benchmark in this report ($r=0.658$, AUC 0.609/0.680) was superseded by the more reliable held-out figures Data Science provided ($r=0.642$, AUC 0.596/0.662); this cross-track dependency is now closed.

10. Key findings / validation outcomes

The Week 6 risk score and driver findings are substantively correct and now more rigorously validated: reproducible, segment-stable, and directionally consistent with an independent model. One headline figure required correction. The reminder effect not varying by risk tier is now a confirmed, adequately powered result.

11. Major challenges encountered

Diagnosing the root cause of the 59.0% vs. 56.0% discrepancy required re-deriving the calculation rather than just re-running it, since the original code was not available for that specific summary figure — it had to be reconstructed from the report text. Data Science's model output was not available at the start of this cycle, so an interim proxy model was used first and later replaced once their actual results arrived.

12. Important decisions made

Decided to correct and use the pooled (56.0%) rate going forward rather than the originally reported figure; used an interim in-house model for cross-track validation while waiting on Data Science, then replaced it with their actual model once received, and adopted their held-out AUC figures as the more reliable reference over this report's original in-sample figures.

13. Remaining limitations

Dataset remains fictional/synthetic; findings are observational (association, not causation); this report's original in-sample benchmark is retained for transparency but superseded by Data Science's held-out figures; risk-score thresholds remain heuristic; the reminder-timing/channel A/B test is designed but not yet run.

14. Remaining issues or dependencies

The minor non-monotonic dip in the 25-34 age group should be monitored on updated data; the stratified reminder-timing test needs to be executed in Week 8; the held-out AUC benchmark should be propagated to any other project document that still cites the original in-sample figures.

15. Contribution to the overall HealthConnect project

This testing cycle gives the HC-POD a statistically validated, segment-stable, and now correctly reported risk-triage tool, plus adequately powered evidence that reminders should stay universal while additional support is targeted by risk tier — and, with Data Science's results incorporated, a closed-loop cross-track validation ready for Week 8 final integration.

16. What must be completed before Week 8

Design and, where feasible, execute the stratified reminder-timing/channel test; propagate the corrected 56.0% figure and the held-out AUC benchmark (0.596/0.662) to any other project document or dashboard; coordinate with Data Science and ML Engineering on how the rule-based score and the refined model will co-exist operationally.